

Friday 09/04/2026: Counterexample

CSCI 246: Discrete Structures

Textbook reference: Sec. 6, Scheinerman

Today's Agenda

- Weekly quiz (15 mins)
- Review previous group exercises ($\approx$ 15 mins)
- Mini-lecture ($\approx$ 5 mins)
- Group exercises ($\approx$ 15 mins)

Quiz Set up

- **Sheet of paper:** Please bring your own sheet of paper to class each day for quizzes if possible. However, if you don't have any, you are welcome to take a blank sheet of paper from the stack in the front of the room.
- **Write your last name on the back:** Please write your first and last name on the front of the page where you will do your work. Then, on the BACK of the page, please write your last name in large letters. This will help us return the graded quizzes efficiently.
- **Rules of conduct:** For all quizzes in the course, you should use only paper and pencil. Please close your computers and textbooks, and put away your cellphones.

**Question: How do you give a counterexample
to
“A implies B”?**

Unidirectional implication: Counterexample

Recall. Here's the truth table for unidirectional implication.

A	B	If A, then B $A \Rightarrow B$
T	T	T
T	F	F
F	T	T
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Unidirectional implication: Counterexample

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Proposition. If you live in Montana, then you live in Bozeman.

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$\underbrace{\hspace{10em}}_A \quad \underbrace{\hspace{10em}}_B$

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Counterexample.

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So we want to try to find a case where the antecedent (A) holds, but the consequent (B) is false. Let's try to apply this.

Proposition. If $\underbrace{\text{you live in Montana}}_A$, then $\underbrace{\text{you live in Bozeman}}_B$.

Counterexample. Matt lives in Helena. So he lives in Montana (A is true) but not in Bozeman (B is False).

**Question: How do you give a counterexample
to
“A if and only if B”?**

Bidirectional implication: Counterexample

Recall. Here's the truth table for bidirectional implication.

A	B	A if and only if B
		$A \iff B$
T	T	T
T	F	F
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So we want to try to find a case where the truth values for **A** and **B** don't match. Let's try to apply this.

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So we want to try to find a case where the truth values for **A** and **B** don't match. Let's try to apply this.

Proposition. A state is Montana if and only if it borders Canada.

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So we want to try to find a case where the truth values for **A** and **B** don't match. Let's try to apply this.

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Proposition. $\underbrace{\text{A state is Montana}}_A$ if and only if $\underbrace{\text{it borders Canada}}_B$.

Counterexample. Michigan borders Canada (**B** is true) but it is not Montana (**A** is False).